

Ali Ataollahi

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EDUCATION

B.Sc. Computer Engineering

2020 - 2025

Department of ECE, University of Tehran, Tehran, Iran

Cumulative GPA: 17.88/20 (3.84/4.00)

High School Diploma

2017 - 2020

Shahid Soltani 4 High School, Karaj, Iran

GPA: 19.66/20 (4.00/4.00)

INTERESTS

- Formal Verification
- Model Checking
- Cyber Physical Systems
- Program Analysis
- Compiler Design
- Theorem Provers

PUBLICATIONS

Extending Moxichecker

Salih Ates, Dirk Beyer, Po-Chun Chien, **Ali Ataollahi**, Marziyeh Mousavi, Nian-Ze Lee
in preparation

Smart Contracts Verification using mCRL2

Nazanin Torabi Goudarzi, Fatemeh Ghassemi, **Ali Ataollahi**, Hossein Hojjat
in preparation

Adaptive Programming

Ali Ataollahi, Fatemeh Ghassemi, Eduard Kamburjan, Marjan Sirjani
in preparation

Reachability analysis of Hybrid Rebeca models

Saeed Zhiany, Fatemeh Ghassemi, Nesa Abbasimoghadam, Ali Hodaei, **Ali Ataollahi**, József Kovács, Erika Ábrahám, Marjan Sirjani
published in Journal of Systems Architecture, available at [this link](#)

Generalizing Featured Finite State Machines Constructed by Model Merging

Shaghayegh Tavassoli, Ramtin Khosravi, **Ali Ataollahi**
submitted to SoSyM, preprint available at [this link](#)

RESEARCH EXPERIENCE

Ludwig Maximilian University of Munich, SoSy-Lab

- **Moxichecker Project**: Developed symbolic model checking algorithms to verify safety in hardware and cyber-physical systems. These algorithms use mathematical analysis to check system behavior and ensure critical components work reliably. I adapted these methods for large-scale circuit designs, creating tools that find design errors before deployment.

University of Tehran, Formal Methods and System Verification Lab

- **Hybrid Rebeca Project**: Focused on verifying asynchronous, event-based Cyber-Physical Systems (CPS) through model checking techniques. Contributed to developing a tool that generates the state space for systems specified by Hybrid Rebeca, a formal, actor-based modeling language designed for asynchronous,

event-driven distributed systems. Hybrid Rebeca extends Rebeca by incorporating continuous variables and dynamic equations to capture the physical behaviors of CPS accurately.

- **Smart Contract Verification:** As a research assistant, I focused on developing a formal verification method for smart contracts, aiming to ensure code correctness and prevent security vulnerabilities. Model checking techniques were used to detect potential attacks by extracting abstract models from Solidity code, discarding elements irrelevant to specific vulnerabilities. The verification process involved testing these models against predefined formulae that identify symptoms of known attack patterns, effectively pinpointing security weaknesses.

University of Tehran, Dependable Systems Design Lab

- **Real-Time Object Detection:** Presented an optimized approach for real-time object detection in embedded CPS using YOLOv8 with a dynamic deadline. Given limited computational power and memory in CPS environments, our method adjusted processing deadlines based on available resources, ensuring timely and accurate detection. We employed optimization techniques—including pruning, quantization, layer skipping, dynamic path selection, and multi-head detection—to balance speed and accuracy.

Remote Collaboration

- **Soren:** Developed a Joern-based static analysis plugin for API attack surface discovery in Java microservices. Enhanced CPG analysis to handle annotations and reflection through custom data-flow techniques. Built a pipeline converting code property graphs into OpenAPI specifications for endpoint visualization. Implemented Docker-based Joern services with JSON output for workflow integration.

COURSERA ONLINE COURSES

System Validation: Automata and Behavioural Equivalences

Eindhoven University of Technology | Instructor: **Jan Friso Groote**

Automated Reasoning: Symbolic Model Checking

Eindhoven University of Technology | Instructor: **Hans Zantema**

Automated Reasoning: Satisfiability

Eindhoven University of Technology | Instructor: **Hans Zantema**

Blockchain Specialization: Smart Contracts

University at Buffalo | Instructor: **Bina Ramamurthy**

Programming Languages

Korea Advanced Institute of Science and Technology(KAIST) | Instructor: **Sukyong Ryu**

TEACHING EXPERIENCE

Compiler Design Tutorial Series

2024–Present

YouTube Channel

TEACHING ASSISTANT EXPERIENCE

Compiler and Programming Languages Design

Spring 2024, Fall 2024

Prof. Ghassemi (*Fall*), **Dr. S. Tavassoli** (*Spring*)

Head Teaching Assistant

Database Design

Spring 2024, Fall 2024

Dr. A. Shakery

Advanced Programming

Spring 2022 - 2024

Prof. Khosravi

Data Structures

Spring 2022 - Fall 2024

Dr. H. Faili

Engineering Statistics

Fall 2022 - 2023

Dr. B. Bahrak, Dr. M. Tavassolipour

Computer Networks
Dr. A. Khonsari

Fall 2023

Computer Aided Design
Dr. M. Ersali, Prof. M. Modarressi

Fall 2023

Operating System
Dr. M. Kargahi

Fall 2023 - 2024

Assignment Designer and Lecturer Assistant

Operations Research
Dr. A. Ramezani Moghadam

Fall 2023

PROJECTS

Formal Verification with Isabelle/HOL Applied formal methods to verify properties of hybrid and dynamic systems using Isabelle/HOL, focusing on correctness, safety, and automated proof strategies.

Baloot Project This project involved testing an e-commerce system built with Spring Boot and a RESTful API, applying various testing methodologies: JUnit unit tests, Mockito mocking, JaCoCo coverage, PITest mutation testing, and Cucumber behavior-driven testing.

IoT-Based Access Control System Developed a real-time door access system with RFID authentication to manage secure entry, validate employee IDs, and log access attempts. Led server-side development in C++ and Qt, implementing HTTP and WebSocket servers for hardware-GUI communication.

Motion-Based Authentication System An Android app for user authentication using motion sensors, utilizing accelerometer and gyroscope data to identify unique movement patterns. Implemented algorithms for pattern matching to enhance security through device-based biometrics.

XV6 Modernized This project customized the MIT xv6 operating system with enhancements in the CLI, scheduling, and system calls to enable functionality comparable to modern Linux-based systems.

Socket-Based Hotel Management Hotel management system with server and multiple clients communicating via sockets, processing requests and managing reservation database.

UT Trade Language Compiler A compiler project for algorithmic cryptocurrency trading, featuring four development phases: lexical and syntax analysis for an LL(K) parser, AST creation, and optimized Java bytecode generation with short-circuit evaluation and stack management.

FunctionCraft Language Compiler As a Teaching Assistant, I developed a Java-based compiler for the FunctionCraft language. Using ANTLR4, the compiler performs lexical analysis, AST generation, and JVM bytecode generation, supporting key functional programming.

NLP Course Projects A collection of projects covering key NLP tasks, including Semantic Role Labeling, fine-tuning LLaMA/RoBERTa, and developing chatbots with LangChain.

HONORS AND AWARDS

- Among top 15% of B.Sc. students in Computer Engineering at University of Tehran.
- Ranked 89th (top 0.1%) among almost 165,000 participants in the Nationwide Iranian University Entrance Exam in Mathematics and Physics.

SKILLS

- **Programming Languages:**
 - Experienced in **C**, **C++**, **Python**, **Java**, **Scala**. Familiar with **Go**, **Bash**, **Solidity**, **mCRL2**.
- **Tools:** ANTLR, Z3, Isabelle, Coq, Lark, Graphviz, L^AT_EX